

Project Title

Data analysis with house sales data.

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1. Abstract

This project presents a complete data analysis workflow using a housing sales dataset from India. The data contains structural features like bedrooms, bathrooms, and living area to help predict property prices. First, data quality checks were performed to inspect the dataset for duplicates and missing values. Next, unnecessary columns like IDs and geographical coordinates were dropped to clean the data. A correlation analysis was then used to find the features that affect house prices the most. Using these top features, a supervised Linear Regression model was trained to predict property prices. The trained model successfully achieved an R-squared accuracy score of 0.5629. Finally, an unsupervised K-Means clustering model was built with 10 target groups and evaluated using an F1-score and a confusion matrix.

2. Introduction

In real estate analytics, processing multi-dimensional datasets is essential to understand property market values. This project focuses on analyzing housing information to discover how structural characteristics influence final house prices. The entire project was built using Python within a Google Colab notebook, utilizing key libraries like pandas, numpy, and scikit-learn.

During the first two weeks of the internship, training was provided on the following core topics:

- Python programming paradigms for data engineering.
- Exploratory Data Analysis (EDA) and data profiling.
- Data quality assurance, data cleaning, and handling missing values.
- Introduction to machine learning models and validation strategies.

3. Project Objective

- To implement a data-cleaning pipeline to remove non-predictive features.
- To identify key structural drivers of house prices using correlation metrics.
- To build a Linear Regression model to predict property prices.
- To implement an unsupervised K-Means model to group similar properties.

4. Methodology

The project was executed in clear, structured steps:

- **Data Ingestion & Cleaning:** The dataset `house_price_india.csv` was loaded into pandas. Unnecessary columns (`id`, `Date`, `Latitude`, `Longitude`, `Renovation Year`, `Postal Code`, `living_area_renov`, `lot_area_renov`) were dropped to clean the data.
- **Feature Selection:** A correlation matrix was calculated using `.corr()`. Features with a strong correlation with price greater than 0.5 were selected: number of

bathrooms, living area, grade of the house, and Area of the house(excluding basement).

- **Data Splitting:** The dataset was split into input features (X) and the target price (y). To validate our model, the data was split into **70% for training data** and **30% for testing data** using a fixed random_state of 123.
- **Model Training & Clustering:** A Linear Regression model was trained on the 70% training data. Additionally, a K-Means clustering model was initialized with n_clusters=10 to group the dataset into 10 distinct property categories.

5. Data Analysis and Results

- **Data Quality Result:** The dataset was found to be clean with zero missing values and zero duplicate rows.
- **Regression Model Output:** The Linear Regression model achieved an **R-squared (R^2) score of 0.56289**. This means our 4 chosen features explain 56.29% of the variance in house prices.
- **Clustering Output:** The 10-cluster K-Means model generated an array of cluster labels for the houses. The macro-averaged F1-score was 9.349e-05, and printing the confusion_matrix showed a sparse grid of corners due to the continuous nature of price_data:

```
Question 13: Confusion Matrix for Clustering (3 Marks)

Generate and display the confusion matrix comparing your mapped cluster labels (from Question 9) to the true labels ( y ). Use confusion_matrix from sklearn.metrics . This helps evaluate how well your clusters align with the actual digit classes without requiring visual plotting.

Expected Output: A 10x10 confusion matrix printed as an array or a Pandas DataFrame.

[ ] from sklearn.metrics import confusion_matrix
    matrix_output = confusion_matrix(y,mapped_labels)
    print(matrix_output)

... [[0 0 0 ... 0 0 0]
     [0 0 0 ... 0 0 0]
     [0 0 0 ... 0 0 0]
     ...
     [0 0 0 ... 0 0 0]
     [0 0 0 ... 0 0 0]
     [0 0 0 ... 0 0 0]]
```

6. Conclusion

The project successfully demonstrated how data cleaning and machine learning can be used to analyze real estate prices. The analysis proved that living area and the house grade are major factors in determining property costs. While the baseline linear model performed well by capturing 56.29% of price variations, future work can explore advanced non-linear models to improve prediction accuracy further.

7. APPENDICES

1. References : Official documentation for Python libraries (pandas, scikit-learn).
2. Github_link:<https://github.com/ananyak0501-cyber/IDEAS-TIH-Summer-Internship-2026.git>

3. Document Link : The notebook file and house_price_india.csv dataset are uploaded together inside the project folder on GitHub